Differential Equations: Workshop

Compiled by Maximus Pung Jun

Total: 54 marks. Give non-exact numerical answers correct to 3 significant figures unless otherwise stated.

Questions

1 [MI/2023 Prelim/P2/Q1]

(i) It is given that $x\dfrac{\mathrm{d}y}{\mathrm{d}x}=x^3+xy+2y,\ x\neq 0$. Using the substitution $y=vx^2$, show that the differential equation can be reduced to

$$\frac{\mathrm{d}v}{\mathrm{d}x}=v+1.$$

[2]

(ii) Given that $y=1$ when $x=1$, solve the differential equation

$$x\frac{\mathrm{d}y}{\mathrm{d}x}=x^3+xy+2y.$$

[3]

2 [YIJC/2024 Prelim/P1/Q4]

(a) It is given that $x\dfrac{\mathrm{d}y}{\mathrm{d}x}=xy\left(\ln x+\ln y\right)-y$. Using the substitution $w=xy$, show that the differential equation can be transformed to $\dfrac{\mathrm{d}w}{\mathrm{d}x}=\mathrm{f}(w)$, where the function $\mathrm{f}(w)$ is to be found. **[3]**

(b) Hence, given that $y=\dfrac{1}{2}\mathrm{e}^3$ when $x=2$, solve the differential equation $x\dfrac{\mathrm{d}y}{\mathrm{d}x}=xy\left(\ln x+\ln y\right)-y$, to find y in terms of x. **[5]**

3 [JPJC/2025 Prelim/P2/Q1]

To prepare for the General Election, a political party studies the effectiveness of spreading information on Instantglam, a social media platform, by uploading a particular video. The number of Instantglam users who have viewed the video increases at a rate that is directly proportional to the number of Instantglam users who have yet to view the video. The number of Instantglam users who have viewed the video t hours after the video is uploaded is x. The total number of Instantglam users is P, assumed to be fixed.

(i) Write down a differential equation and show that $x=P\left(1-\mathrm{e}^{-kt}\right)$, where k is a positive constant. **[4]**

It is known that 12 hours after the video is uploaded, half the total number of Instantglam users have viewed the video.

(ii) Estimate to the nearest hour, the time needed for 80% of the total number of Instantglam users to have viewed the video. **[3]**

(iii) Sketch the graph of x against t. **[2]**

4 [RVHS/2025 Prelim/P1/Q13]

A cafe roasts its own coffee beans and packages them to be sold. The amount of roasted coffee beans remaining in the cafe at time t days is denoted by x kg. The cafe produces roasted coffee beans at a fixed rate of 10 kg/day, and sells the roasted coffee beans at a rate proportional to x^2 kg/day.

(a) Given that there is no change in the amount of roasted coffee beans remaining in the cafe when there is 5 kg of beans remaining, show that $\frac{\mathrm{d}x}{\mathrm{d}t} = 10 - \frac{2}{5}x^2$. Find the general solution of this differential equation. (You do not have to make x the subject.) **[3]**

Due to improvement in roasting capabilities of the cafe, it is now able to produce roasted coffee beans at the rate of $3x$ kg/day, and sells the roasted coffee beans at a rate proportional to x^2 kg/day.

(b) Given that amount of roasted coffee beans is increasing at the rate of 6 kg/day when the amount of roasted coffee beans remaining in the cafe is 6 kg, write down the modified differential equation relating $\frac{\mathrm{d}x}{\mathrm{d}t}$ and x. **[2]**

(c) Given that the initial amount of roasted coffee beans in the cafe is 13.5 kg, solve the differential equation in part (b), expressing x in terms of t.

Deduce the amount of roasted coffee beans remaining in the cafe in the long run, and sketch the curve of x against t. **[6]**

(d) Based on the curve in part (c), explain in context why this might be favourable for the cafe owner. **[1]**

5 [DHS/2019 Prelim/P1/Q12]

In a particular chemical reaction, every 2 grams of U and 1 gram of V are combined and converted to form 3 grams of W. Let u, v and w denote the mass (in grams) of U, V and W respectively present at time t (in minutes). According to the law of mass action, the rate of change of w with respect to t is proportional to the product of u and v. Initially, $u = 40$, $v = 50$ and $w = 0$.

(i) Show that $\frac{\mathrm{d}w}{\mathrm{d}t} = k(w-60)(w-150)$, where k is a positive constant. **[3]**

It is observed that when $t = 5$, $w = 10$.

(ii) Find w when $t = 20$, giving your answer to two decimal places. **[7]**

(iii) What happens to w for large values of t? **[2]**

6 [TMJC/2025 Prelim/P2/Q4]

A viral trend on social media gains popularity rapidly but eventually slows down due to oversaturation. Let P represent the popularity score of the trend at time t (measured in days), where $0 \le t \le 100$. The change in the trend's momentum is modelled by

$$\frac{\mathrm{d}^2P}{\mathrm{d}t^2} = -0.05\left(\frac{\mathrm{d}P}{\mathrm{d}t}\right)^2.$$

(a) By substituting $v = \frac{\mathrm{d}P}{\mathrm{d}t}$, show that $v = \frac{20}{t+C}$ where C is a real constant. **[3]**

(b) Given that $P = 0$ and $v = 10$ when $t = 0$, find the particular solution of P in terms of t. **[4]**

(c) Find the popularity score of the trend when $t = 30$. **[1]**

Answers

1 **(i)** Shown.

(ii) $1+\frac{y}{x^2}=2\mathrm{e}^{x-1}$, i.e. $y=x^2\left(2\mathrm{e}^{x-1}-1\right)$

2 **(a)** $\mathrm{f}(w)=w\ln w$

(b) $y=\frac{\mathrm{e}^{3\mathrm{e}^{x-2}}}{x}$

3 **(i)** $\frac{\mathrm{d}x}{\mathrm{d}t}=k(P-x)$; shown.

(ii) $27.9\approx 28$ hours

(iii) Increasing curve from O, approaching $x=P$ (see solutions).

4 **(a)** $\frac{1}{10}\ln\left|\frac{x-5}{x+5}\right|=-\frac{2}{5}t+c$

(b) $\frac{\mathrm{d}x}{\mathrm{d}t}=3x-\frac{1}{3}x^2$

(c) $x=\frac{27\mathrm{e}^{3t}}{3\mathrm{e}^{3t}-1}$; 9 kg in the long run (sketch in solutions)

(d) There is always roasted coffee left to sell, and the stable amount lets the owner plan storage.

5 **(i)** Shown.

(ii) $w=29.32$

(iii) $w\to 60$

6 **(a)** Shown.

(b) $P=20\ln\left(\frac{t+2}{2}\right)$

(c) 55.5

END OF WORKSHEET